Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

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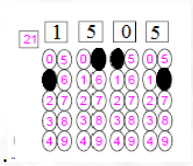
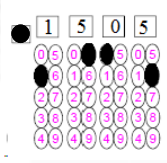
SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 07-01-2026**

Time: 09.00Am to 12.00Pm **GTM-10** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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07-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main_GTM-10_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let A be a matrix of order 3×3, $|A|=5$, *if* $\left|2\,adj\left(3A\,adj\left(2A\right)\right)\right|=2^{\alpha}.3^{\beta}.5^{\gamma},\alpha,\beta,\gamma\in N$ then $\alpha+\beta+\gamma$ is equal to

1) 25 **2)** 26 **3)** 27 **4)** 28

2. If the shortest distance between the lines $\frac{x-\lambda}{-2}=\frac{y-2}{1}=\frac{z-1}{1}$ *and* $\frac{x-\sqrt{3}}{1}=\frac{y-1}{-2}=\frac{z-2}{1}$ is 1. Then the sum of all possible values of λ

1) 0 **2)** $2\sqrt{3}$ **3)** $3\sqrt{3}$ **4)** $-2\sqrt{3}$

3. $\int\frac{(1+x)e^{x}}{\cot\left(xe^{x}\right)}dx$ =

1) $\log\left|\cos\left(xe^{x}\right)\right|+c$ **2)** $\log\left|\cot\left(xe^{x}\right)\right|+c$

3) $\log\left|\sec\left(xe^{-x}\right)\right|+c$ **4)** $\log\left|\sec\left(xe^{x}\right)\right|+c$

4. The area (in sq. units) of the region enclosed between the parabola $y^2=2x-1$ and $y^2=4x-3$ is

1) $\frac{1}{3}$ **2)** $\frac{1}{6}$ **3)** $\frac{2}{3}$ **4)** $\frac{3}{4}$

5. If $\lim\limits_{x\to 0}\left(\cos x+a^3\sin\left(b^6x\right)\right)^{\frac{1}{x}}=e^{512}$, then the value of ab^2 is equal to:

1) -512 **2)** 512 **3)** 8 **4)** $8\sqrt{8}$

6. Let $\vec{\alpha}=3\hat{i}+\hat{j}$ *and* $\vec{\beta}=2\hat{i}-\hat{j}+3\hat{k}$ If $\vec{\beta}=\vec{\beta}_1-\vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ *and* $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta}_1\times\vec{\beta}_2$ is equal to

1) $-3\hat{i}+9\hat{j}+5\hat{k}$ **2)** $3\hat{i}-9\hat{j}-5\hat{k}$

3) $\frac{1}{2}\left(-3\hat{i}+9\hat{j}+5\hat{k}\right)$ **4)** $\frac{1}{2}\left(3\hat{i}-9\hat{j}-5\hat{k}\right)$

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7. Let the hyperbola $H:\frac{x^2}{a^2}-y^2=1$ and the ellipse $E:3x^2+4y^2=12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E. If e_H *and* e_E are the eccentricities of H and E respectively, then the value of $12\left(e_H^2+e_E^2\right)$ is equal to_____

1) 34 **2)** 38 **3)** 40 **4)** 42

8. Let $x=x(t)$ and $y=y(t)$ be solution of the differential equations $\frac{dx}{dt}+ax=0$ and $\frac{dy}{dt}+by=0$ respectively $(a,b\in R)$. Given that $x(0)=2;y(0)=1$ and $3y(1)=2x(1)$, the value of t, for which $x(t)=y(t)$, is:

1) $\log_{\frac{2}{3}}2$ **2)** $\log_4 3$ **3)** $\log_3 4$ **4)** $\log_{\frac{4}{3}}2$

9. For the function $f(x)=(\cos x)-x+1, x\in\mathbb{R}$ between the following two statements

Statement-I: $f(x)=0$ for only one value of x is $[0,\pi]$

Statement-II: $f(x)$ is decreasing in $\left[0,\frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2},\pi\right]$

1) Both Statement-I and Statement-II are correct
2) Only Statement-I is correct
3) Both Statement-I and Statement-II are incorrect
4) Only Statement-II is correct

10. If $\left(1+x+x^2\right)^{48}=a_0+a_1x+a_2x^2+....+a_{96}x^{96}$ then the value of $a_0-a_2+a_4-a_6+.....+a_{96}$ is

1) -1 **2)** 0 **3)** 1 **4)** 48

11. Let $f(x)=\frac{\left(e^x-1\right)^2}{\sin\left(\frac{x}{a}\right)\log\left(1+\frac{x}{4}\right)}$ *for* $x\neq 0$ and f(0) = 12. If f is continuous at x = 0 , then the value f a is equal to

1) 1 **2)** -1 **3)** 2 **4)** 3

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12. The number of ways, in which 5 girls and 7 boys can be seated at a round table so that no two girls sit together is

1) 720 **2)** $126(5!)^2$ **3)** $7(360)^2$ **4)** $7(720)^2$

13. Four distinct points $(2k,3k),(1,0),(0,1)\,and\,(0,0)$ lie on a circle for k equal to :

1) $\frac{2}{13}$ **2)** $\frac{3}{13}$ **3)** $\frac{5}{13}$ **4)** $\frac{1}{13}$

14. If $f(n)=\frac{3^n}{3^n+\sqrt{3}}$ then $\sum_{r=1}^{99} f\left(\frac{r}{100}\right)$ is

1) 99 **2)** 100 **3)** 49.5 **4)** 50

15. The number of real solution(s) of the equation $9^{\log_3(\log_e x)}=\log_e x-(\log_e x)^2+1$ is

1) zero **2)** 1 **3)** 2 **4)** 3

16. Let Z be complex number such that $\left|\frac{z-i}{z+2i}\right|=1\,and\,|z|=\frac{5}{2}$ then the value of $2(|z+3i|)$ is

1) $\sqrt{10}$ **2)** $2\sqrt{3}$ **3)** 7 **4)** 15

17. The equation of the parabola whose focus is $(-3,0)$ and the directrix is $x+5=0$ is

1) $y^2=4(x+5)$ **2)** $y^2=4(x+4)$ **3)** $y^2=4(x+3)$ **4)** $y^2=4(x-3)$

18. Let f(x) be a polynomial of degree 4 such that f(1) = 7 and attains its local minimum value 3 at both x = 2 and x = 3. The local maximum value of f(x) is equal to

1) $\frac{49}{16}$ **2)** $\frac{25}{8}$ **3)** $\frac{5}{2}$ **4)** $\frac{47}{16}$

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19.

COLOUM-I		COLOUM-II	
A)	If the middle term of the A.P. $a_1, a_2, a_3, ..., a_{15}$ is $\frac{1}{5}$ then its sum is equal to	p)	4
B)	The infinite geometric series with common ratio $\frac{1}{y^2-4y+5}$ has a finite sum when y can not take value equal to	Q)	4/3
C)	If $x, x+1, 2x+2, \ldots..$, are in GP then the next term of the sequence is	R)	2
D)	If $K = \frac{1}{2.7} + \frac{1}{7.12} + \frac{1}{12.17} + ...\infty$ then 40K is equal to	S)	8
		T)	3

1) $A \to R, B \to T, C \to P, D \to S$ **2)** $A \to R, B \to S, C \to T, D \to P$

3) $A \to T, B \to R, C \to S, D \to P$ **4)** $A \to S, B \to P, C \to R, D \to T$

20. Let ω be a complex cube roots of unity with $\omega \neq 1$. A fair dice is thrown three times . If r_1, r_2 *and* r_3 are the numbers obtained on the die, then probability that $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0$ is

1) $\frac{1}{18}$ **2)** $\frac{1}{36}$ **3)** $\frac{2}{9}$ **4)** $\frac{1}{9}$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. The number of real roots of the equation $x^2 + 5|x| + 4 = 0$ is equal to

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22. Consider 10 observations $x_1, x_2, ...x_{10}$, such that $\sum_{i=1}^{10}(x_i-\alpha)=2$ and $\sum_{i=1}^{10}(x_i-\beta)^2=40$,

Where α, β are positive integers. Let the mean and variance of the observations be $\frac{6}{5}$

and $\frac{84}{25}$ respectively. Then β/α is equal to:

23. Given $a,b \in \{0,1,2,.....6\}$ consider the system of equation

$x+y+z=4, 2x+y+3z=6, x+2y+az=b$

If the number of ordered pairs(a,b) so that the system of the equation has unique solution

is λ, then $\frac{\lambda}{7}$ equals

24. $f(x)=\begin{cases} 2x^2+3 & ; x>3 \\ ax^2+bx+12 & ; x\le 3 \end{cases}$ is differentiable everywhere, then $|a+b|$ is equal to

25. If f(x) is defined on [-3,3] by $f(x)=4x^3-2x+1$ and $g(x)=\frac{f(x)-f(-x)}{|x|+2}$, then the

value of $\int_{-3}^{3} g(x)$ is equal to

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. The intensity of radiation emitted by the sun has its maximum value at a wavelength of 510 nm and that emitted by the north star has the maximum value at 350 nm. If these stars behave like black bodies, then the ratio of the surface temperature of the sun and north star is

1) 1.46 **2)** 0.69 **3)** 1.21 **4)** 0.83

27. In a photoelectric experiment, the wavelength of the light incident on a metal is changed from 300 nm to 400 nm. The decrease in the stopping potential is close to :

$\left(\frac{hc}{e} = 1240\,\text{nm} - \text{V}\right)$

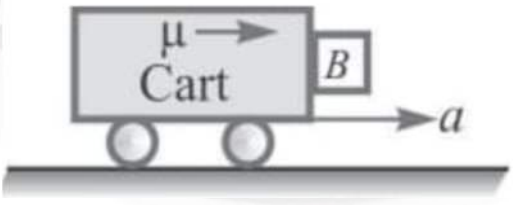
1) 1.5 V **2)** 0.5 V **3)** 2.0 V **4)** 1.0 V

28. A plane e.m. wave of frequency 30 MHz travels in free space along X-direction. The electric field component of the wave at a particular point of space and time is E = 6 V/m along Y-direction. Its magnetic field component B at this point would be:

1) $2\times10^{-8}\,T$ along Z-direction **2)** $6\times10^{-8}\,T$ along X-direction

3) $2\times10^{-8}\,T$ along Y-direction **4)** $6\times10^{-8}\,T$ along Z-direction

29. If μ be the coefficient of friction between the block and the cart, horizontal acceleration of the cart that is required to prevent block B from falling is:



1) μ / g **2)** g / μ **3)** g **4)** $\left(\mu^2 + 1\right)g$

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30. A solid sphere of radius R is charged uniformly through out the volume. At what distance from its surface is the electrostatic potential half of the potential at the centre.

1) R **2)** R/2 **3)** R/3 **4)** 2R

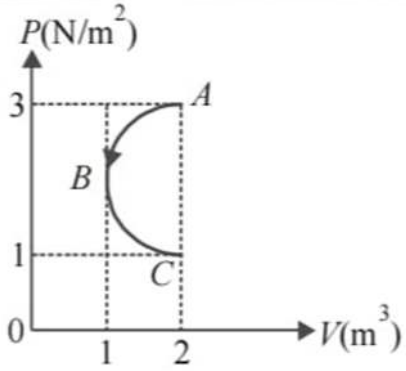
31. A particle is executing simple harmonic motion(SHM) of amplitude A, along the x-axis, about x=0. When its potential energy(PE) equals kinetic energy (KE), the position of the particle will be

1) A **2)** $\frac{A}{\sqrt{2}}$ **3)** $\frac{A}{2\sqrt{2}}$ **4)** $\frac{A}{2}$

32. Magnetic flux (in weber) in a closed circuit of resistance 10 Ω varies with time t (in seconds) according to the equation $\phi = 6t^2 - 5t + 1$. What is the magnitude of the induced current at t = 0.25 seconds?

1) 1.2 A **2)** 0.8 A **3)** 0.6 A **4)** 0.2 A

33. In P–V diagram shown in figure ABC is a semicircle. The work done in the process ABC is



1) 4J **2)** $\frac{-\pi}{2}J$ **3)** $\frac{\pi}{2}J$ **4)** Zero

34. A thin lens made of glass of refractive index μ = 1.5 has a focal length equal to 12 cm in air. It is immersed in a liquid of refractive index $\left(\mu = \frac{4}{3}\right)$ then its new focal length is

1) 48 cm **2)** 36 cm **3)** 24 cm **4)** 12 cm

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35. A uniform thin rod of length L and mass M is rotated about an axis perpendicular to the rod and passing through its center. The moment of inertia is:

1) $\frac{1}{12}ML^2$ **2)** $\frac{1}{3}ML^2$ **3)** $\frac{1}{2}ML^2$ **4)** ML^2

36. A particle of charge q and mass m moves in a circular orbit of radius r with angular speed ω The ratio of the magnitude of its magnetic moment to that of its angular momentum depends on

1) ω and q **2)** ω ,q and m **3)** q and m **4)** ω and m

37. The nuclear radius is given by $R = r_0 A^{1/3}$, where r_0 is constant and A is the atomic mass number. Then,

1) The nuclear mass density of U^{238} is twice that of Sn^{119}

2) The nuclear mass density of U^{238} is thrice that of Sn^{119}

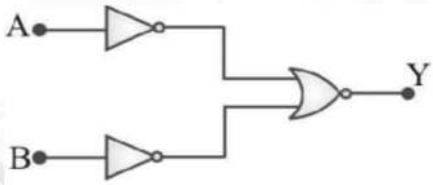
3) The nuclear mass density of U^{238} is the same as that of Sn^{119}

4) The nuclear mass density of U^{238} is half that of Sn^{119}

38. Two block of masses 10 kg and 4 kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse give a velocity of 14 m/s to the heavier block in the direction of the lighter block. The velocity of centre of mass is :

1) 30 m/s **2)** 20 m/s **3)** 10 m/s **4)** 5 m/s

39. Which logic gate is represented by the following combination of logic gate



1) OR **2)** NAND **3)** AND **4)** NOR

40. Two coherent sources produce waves of different intensities which interfere. After interference, the ratio of the maximum intensity to the minimum intensity is 16. The intensity of the waves are in the ratio

1) 16 : 9 **2)** 25 : 9 **3)** 4 : 1 **4)** 5 : 3

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41. **Statement-1:** In a hydrogen atom energy of emitted photon corresponding to transition from n = 2 to n = 1 is greater as compared to transition from n → ∞ to n = 2.

Statement-2: Wavelength of photon is directly proportional to the energy of emitted photon.

1) Both statements are true and Statement 2 is the correct explanation of Statement 1.

2) Both statements are true but Statement 2 is not the correct explanation of Statement 1.

3) Statement 1 is true and Statement 2 is false.

4) Statement 1 is false and Statement 2 is true.

42. **Statement-1:** A parallel plate capacitor is connected across battery through a key. A dielectric slab of dielectric constant k is introduced between the plates. The energy stored becomes k times.

Statement-2: The surface density of charge on the plate remains constant.

1) Both the statements are true and the Statement-2 is the correct explanation of Statement-1

2) Both the statements are true but the Statement-2 is not the correct explanation of Statement-1

3) Statement-1 true but Statement-2 is false

4) Statement-1 is false but Statement-2 is true

43. **Assertion (A)** : The fundamental frequency of an open organ pipe is twice that of a closed organ pipe of the same length

Reason (R) : In an open pipe, both ends act as antinodes, while in a close pipe, one end is a node, and the other is an antinode

1) Both A and R are true, and R is the correct explanation of A

2) A is false, but R is true

3) Both A and R are true, but R is not the correct explanation of A

4) A is true, but R is false

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44.

Column–I (Graph type)		Column–II (Nature of motion)	
A)	Velocity–time graph is a straight line with positive slope	P)	Uniform motion
B)	Displacement–time graph is a straight line	Q)	Uniform acceleration
C)	Velocity–time graph is a horizontal line	R)	Zero acceleration but non-zero velocity
D)	Displacement–time graph is a curve bending upwards	s)	Increasing speed with time (acceleration present)

1) $A \to R, B \to Q, C \to P, D \to S$ **2)** $A \to R, B \to S, C \to S, D \to P$

3) $A \to Q, B \to P, C \to R, D \to S$ **4)** $A \to Q, B \to P, C \to S, D \to R$

45. **Assertion (A):** In a purely inductive AC circuit, the average power consumed over one complete cycle is zero.

Reason (R): In a purely inductive circuit, the current lags behind the applied voltage by a phase angle of 90^0

1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

2) (A) is false but (R) is true

3) Both (A) and (R) are true and (R) in correct explanation of A

4) A is true and (R) is false

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

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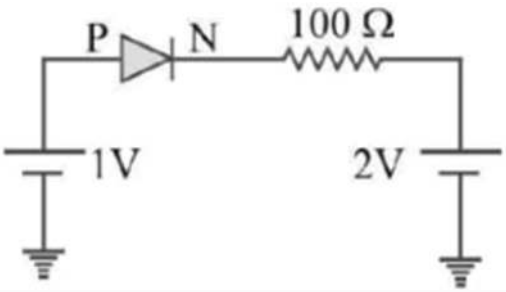
46. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is 54 μT. What will be its value in μT at the centre of the loop

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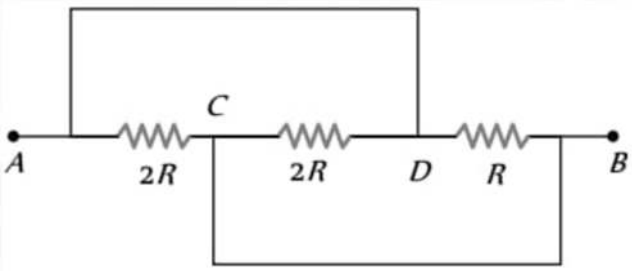
47. The current through an ideal PN-junction shown in the following circuit diagram will be



48. Water flows in a horizontal pipe with speeds 3 m/s and 6 m/s at two points. The pressure difference (in Pa) between these two points is 100p find the value of p $\left(\rho = 1000 kg/m^3\right)$

49. The dimensional of gravitation constant G is $\left[M^a L^b T^c\right]$. Find $a+b+c$.

50. What is the equivalent resistance between A and B $\left(R = 10\Omega\right)$



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JEE Advanced 2023 VAVILALA CHAITANYA REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

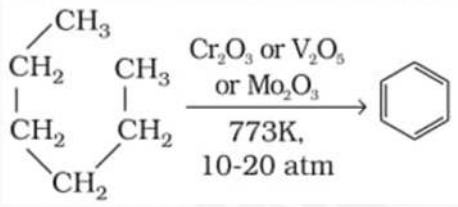
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Which of the following Reactions are correct

(i) $2CH_4(g)+O_2(g) \xrightarrow[100atm]{Cu,523K} 2CH_3OH$

(ii) $CH_4+O_2 \xrightarrow[\Delta]{Mo_2O_3} H-\overset{\overset{O}{\|}}{C}-H+H_2O$

(iii) $2C_2H_6+3O_2 \xrightarrow[(CH_3COO)_2Mn]{\Delta} 2CH_3COOH+2H_2O$

(iv) 

1) i,ii,iii,iv **2)** ii & iii only **3)** i & ii only **4)** i & iii only

52. Which of the following compounds will give silver mirror with ammoniacal silver nitrate solution?

A. Formic acid B. Formaldehyde C. Benzaldehyde D. Acetone

Choose the correct answer from the options given below:

1) B and C only **2)** C and D only **3)** A, B and C only **4)** A only

53. Which of the following is correct (where k is rate constant)

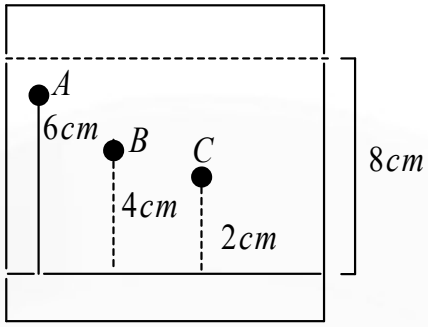
1) $K=\frac{x}{t}$(Zero order) $\left(x=[A]_0-[A]_t\right)$ **2)** $K=\frac{x}{t}$(first order) $\left(x=[A]_0-[A]_t\right)$

3) $K=\frac{1}{t}\ln\frac{[A]_0}{[A]_t}$(Zero order) **4)** $K=\frac{1}{t}\ln\frac{[A]_0}{[A]_t}$(second order)

54. Three organic compounds A, B and C were allowed to run in thin layer chromatography using hexane and gave the following result (see figure). The R_f value of the most polar compound is $X\times10^{-2}$ the value of X is_____

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1) 50 **2)** 25 **3)** 75 **4)** 12.57

55. **Assertion (A) :** The van't Hoff factor of Hg_2I_2 when it 100% ionised is 3.

Reason (R) :It is dissociated to form mercurous (dimer) and iodide ions.

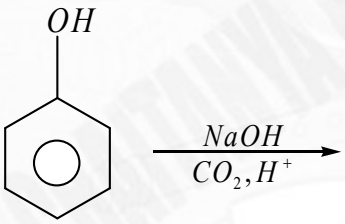
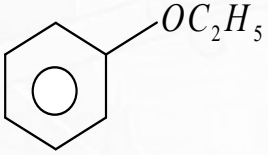
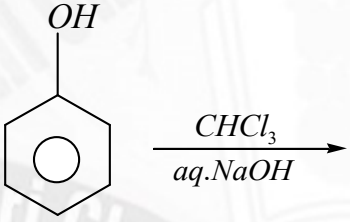
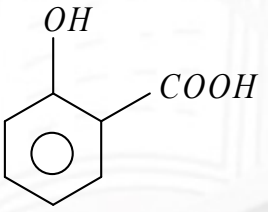
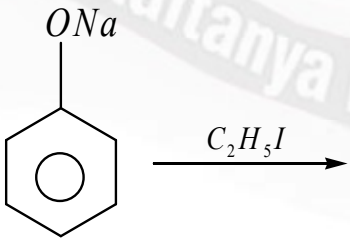
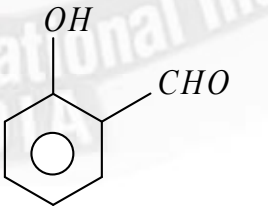
(1) (A) is True, (R) is True; (R) is a correct explanation for (A)

(2) (A) is True, (R) is True; (R) is NOT a correct explanation for (A)

(3) (A) is True, (R) is False

(4) (A) is False, (R) is True

56. Match the following list and choose correct answer

	LIST – I		LIST – II		LIST – III
I)		**1)**		P)	Williamson Synthesis reaction
II)		**2)**		Q)	Reimer Tiemann reaction
III)		**3)**		R)	Kolbe's reaction

1) I – 2 – R **2)** II – 3 – Q **3)** III – 1 – P **4)** All

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57. The element that does not show catenation is

1) C **2)** Ge **3)** Sn **4)** Pb

58. Given below are two statements:

Statement(I): The radii of isoelectronic species increases in the order :

$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$.

Statement(II): The number of lone pair(s) of electrons on central atom of BrF_3 molecule is 2

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both statement (I) and statement (II) are incorrect.

2) statement (I) is correct but statement (II) is incorrect.

3) statement (I) is incorrect but statement (II) is correct.

4) Both statement (I) and statement (II) are correct.

59. Given below are two statements. Once is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The first ionization enthalpy for oxygen is lower than that of nitrogen.

Reason R: The four electrons in $2p$ orbitals of oxygen experience more electron-electron repulsion.

In the light of the above statements, choose the correct answer from the options given below.

1) Both A and R are correct and R is the correct explanation of A

2) Both A and R are correct and R is not the correct explanation of A

3) A is correct but R is not correct

4) A is not correct but R is correct

60. The elemental composition of a compound is $54.2\%C, 9.2\%H$ and $36.6\%O$. If the molar mass of the compound is 132 $g\,mol^{-1}$, the molecular formula of the compound is

[Given: The relative atomic mass of $C:H:O = 12:1:16$]

1) $C_6H_{12}O_3$ **2)** $C_4H_8O_2$ **3)** $C_4H_9O_3$ **4)** $C_6H_{12}O_6$

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61. The correct order of the oxidation states of nitrogen in NO, N_2O, NO_2 and N_2O_3 is

1) $NO_2 < N_2O_3 < NO < N_2O$ **2)** $N_2O < NO < N_2O_3 < NO_2$

3) $N_2O < N_2O_3 < NO < NO_2$ **4)** $NO_2 < NO < N_2O_3 < N_2O$

62. Given below are two statements:

Statement I: H_2Se is more acidic than H_2Te.

Statement II: H_2Se has higher bond enthalpy for dissociation than H_2Te.

In the light of the above statements, choose the correct answer from the options given below:

1) Both statement I and statement II are true.

2) Both statement I and statement II are false.

3) Statement I is true but statement II is false.

4) Statement I is false but statement II is true.

63. The correct order of electron affinity is

1) Cl > F > O **2)** F > O > Cl **3)** F > Cl > O **4)** O > F > Cl

64. Choose the correct option having all the elements with d^{10} electronic configuration from the following.

1) $^{29}Cu, ^{30}Zn, ^{48}Cd, ^{47}Ag$ **2)** $^{46}Pd, ^{28}Ni, ^{26}Fe, ^{24}Cr$

3) $^{28}Ni, ^{24}Cr, ^{26}Fe, ^{29}Cu$ **4)** $^{27}Co, ^{28}Ni, ^{26}Fe, ^{24}Cr$

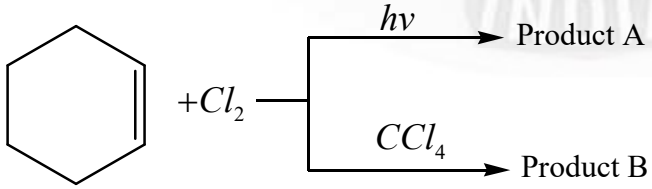
65. In acidic medium, $K_2Cr_2O_7$ shows oxidising action as represented in the half reaction:

$$Cr_2O_7^{2-} + yH^+ + ze^- \rightarrow 2Cr^{+3} + xH_2O$$

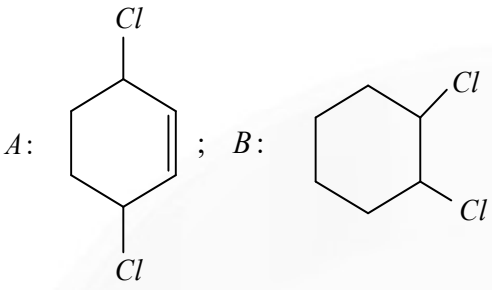
x, y and z respectively are

1) 7, 14, 6 **2)** 14, 7, 6 **3)** 14, 6, 7 **4)** 8, 6, 4

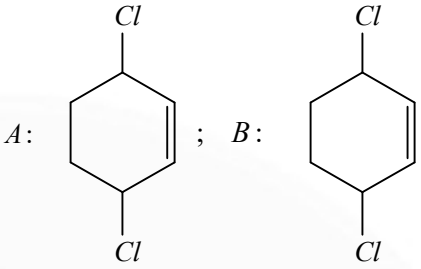
66. Identify product A and product B.



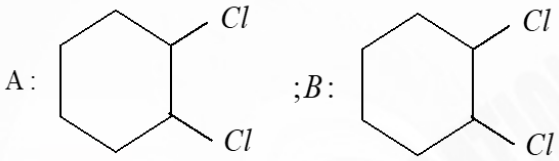
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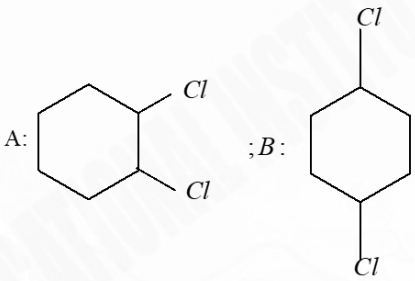
1)



2)



3)



4)

67. The shape of carbocation is

1) Trigonal Planar **2)** Tetrahedral

3) trigonal bipyramidal **4)** Octahedral

68. Given below are two statements:

Statement I: The NH_2 group in aniline is ortho and para directing and a powerful activating group.

Statement II: Aniline does not undergo Friedel-Crafts reaction (alkylation and acylation).

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both Statement I and Statement II are incorrect.

2) Statement I is incorrect but Statement II is correct

3) Both Statement I and Statement II are correct.

4) Statement I is correct but Statement II is incorrect.

69. If one strand of DNA has the sequence ATGCTTCA sequence of the bases in complementary strand is

1) ATGCACT **2)** CATTAGCT **3)** TACGAAGT **4)** GTACTTAC

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70. Formula for Nessler's reagent is

1) $KHgI_3$ **2)** HgI_2 **3)** K_2HgI_4 **4)** KHg_2I_2

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. How many of the following reactions $\Delta H \neq \Delta E$

1) $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$ **2)** $N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$

3) $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ **4)** $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

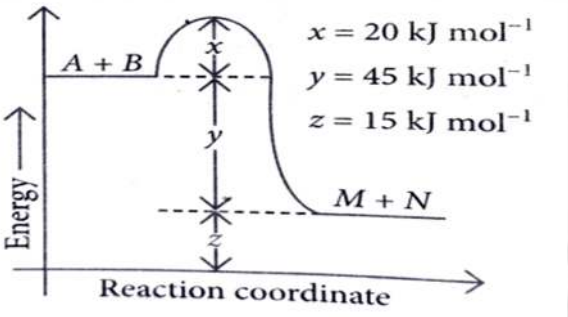
72. How many of the following species are paramagnetic

1) O_2 **2)** B_2 **3)** NO **4)** CO

73. The percentage dissociation of a salt (MX_3) solution at given temperature (van't Hoff factor i = 2) is _____%. (Nearest integer)

74. The quantity of electricity in Faraday needed to reduce 1 mol of MnO_4^- to Mn^{+2} is ___.

75. According to the following figure, the magnitude of the enthalpy changes of the reaction $A + B \rightarrow M + N$ In kJ mol^{-1} is equal to _____ (Integer answer)



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